

A Deep Hybrid Learning Approach For Nail Diseases Classification

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Abstract—In medical domains, the appearance of fingernails can provide clues to underlying systemic diseases or nutritional imbalance; the neglect of such clues could lead to unwanted health complications and less chance for recovery. In this paper, a Deep Hybrid Learning (DHL) approach was proposed to detect nail-based diseases, where a Deep Learning (DL) model is used for feature extraction, and a traditional Machine Learning (ML) classifier is used for classification. The aim is to classify three nail diseases: melanoma, beau's nails, and eczema, in addition to healthy nails. Further, the proposed approach is compared to the transfer learning approach, where a pre-trained model is used for feature extraction and classification. The experiment results indicate that the DHL approach is superior to the transfer learning approach. Specifically, the architecture where the DenseNet201 pre-trained model is used for feature extraction and the SGDClassifier is used for classification, as it achieved an accuracy of 94%.

Keywords—Deep Hybrid Learning, Feature Extraction, Fingernail Disease Prediction, Image Processing.

I. INTRODUCTION

Human nail is formed of keratin, a tough protein that shields the delicate tissues along the border of the fingers or toes. However, its job is more than protecting the sensitive area beneath it. According to some studies, many dangerous diseases may be detected in early stages only by diagnosing nail disorders which are categorized based on whether they affect the nail's shape, texture, or color [1], [2]. Furthermore, nail disorders are a typical reason for dermatologist visits and form about 10% of dermatological conditions [3]. Despite the vital health indications highlighted by the appearance of the nails, nail changes are frequently ignored as long as there is no pain. Although not all nail alterations are harmful, since age might be one of the reasons, certain studies demonstrate a clear link between nail alterations and systemic disorders or nutritional deficiencies [4]. A study was conducted on 157 patients with nail changes (71 males and 86 females) with multiple requirements such as random blood sugar test, skin biopsy, longitudinal nail biopsy, and so on. The ages of participants varied between 1-86 years, and most patients

reported involvement of 2-5 nails. Onychomycosis was the most prevalent nail disease found in individuals with no associated dermatoses. Moreover, some nail diseases showed a clear relation to serious systemic diseases. For instance, clubbing nail disease was linked to Pulmonary Koch's, and chronic obstructive pulmonary disease, while beau's lines appeared in some patients that suffered from typhoid and pneumonia infections. Furthermore, Koilonychia nail disease was observed in patients with iron deficiency [5]. The findings of the study highlight the link between nail diseases and other congenital, developmental, dermatologic, and systemic problems and emphasize the importance of diagnosing nail diseases at early stages to avoid the risk of exacerbations. The color of a healthy human nail is pink, with no unusual dents or cracks [6]. During clinical inspections, doctors typically diagnose the disease by observing the patient's nail color and texture. However, human eyes have limitations in color analysis and cannot capture slight color changes. In addition, some people suffer from color blindness, which makes it impossible to detect such changes. Consequently, identifying the disease may take too long, leading to undesirable health complications and less chance for recovery. Therefore, analyzing nail abnormalities using Machine Learning (ML) and Deep Learning (DL) techniques is better than relying on human sight [1], [7].

In this study, the proposed system uses a model that focuses on detecting three fingernail diseases, namely beau's lines, melanoma, and eczema, in addition to healthy nails. In more detail, a Deep Hybrid Learning (DHL) approach consisting of a combination of DL and ML techniques is used to analyze and extract the features of the fingernail image and then classify it as one of the four classes. The proposed system will provide a reliable, fast, and free nail disease prediction tool accessible anytime and anywhere.

The remaining sections of the paper are organized as follow. Section II explores studies that implemented ML and DL techniques to predict nail diseases. Section III discusses the methods and techniques used for the implementation and the theories and principles behind them. Also, a detailed

description of the dataset is provided. Section IV explains the architecture of the proposed DHL model in depth. Section V demonstrates the performance evaluation metrics used to evaluate the models. Section VI presents the experiment's results and compares the transfer learning approach with the proposed DHL approach. Finally, section VII contains the conclusion and recommendations for future work.

II. LITERATURE REVIEW

Many researchers are utilizing Artificial Intelligence techniques such as imaging processing and ML algorithms to identify human body disorders. This section summarizes some of the studies related to nail disease prediction using AI techniques.

For example, in India, Solapur, Indi et al. [8] proposed a model that categorizes the input fingernail images according to their color. Around 50 various shades of each color were taken into consideration. A digital camera takes the picture, after which the Region of Interest (ROI) is chosen, and the average RGB value is computed. The C4.5 algorithm was used to construct a decision tree and make predictions. Local hospitals in India provided the dataset for training. The model's average accuracy in identifying four potential systemic disorders based on nail color was 65%. The same result was achieved by Sharine et al. [9] from Chennai, India, who presented a system that uses a camera to take a picture of a person's nail and upload that image to the system. Then the system manually selects the region of interest from the nail area.

The chosen area is then further processed to extract nail features like nail color and nail pattern. Next, the convolutional neural network will categorize and process the diseases that can be distinguished from one another based on the color and texture of the nails. Finally, a training dataset and a feature extracted from the input nail image were compared to obtain a result. Weka was used to prepare the dataset. In the experiment, it was discovered that an average of 65% of the results accurately matched the training dataset when using the nail image's color feature. Other researchers identify more diseases, such as In Roorkee, India, Nijhawan et al. [10] used an ensemble of CNNs Architecture to predict 11 nail diseases. For starters, fine-tuning is performed using pre-trained Alex Net architecture. Moreover, four scenarios were considered to assess accuracy.

In the scenarios, the researchers experiment with the images by splitting them into different halves or submitting them as a whole to one or more CNNs and then combining the results in a single feature vector used later for classification. The researchers built the dataset with 4190 total images from eleven different classes. However, the source is not mentioned. Within the four scenarios, the third scenario, where the image is split into two halves that are submitted to two different CNNs along with the whole image submitted to another CNN, demonstrated the best performance in terms of time, which is 4 hours less than the other scenarios and an accuracy of 84.58%. Additionally, in Tamil Nadu, India, Maniyan and Shivakumar [11] proposed the NIPS-McS system, which extracts 13 human nail characteristics and uses them to predict diseases. The back of the palm, which is photographed by a camera, serves as the system's input, then the nail region will automatically be segmented after determining the ROI. Next, the nail's characteristics, such as nail shape, color, and texture, are extracted from the

segmented nail region, after that all the features that have been extracted are used to create a feature vector, which will be compared to the datasets and will be used to predict the diseases.

By collecting the nail shape, color, and texture characteristics and employing the classification algorithm multiclass Support Vector Machine (SVM) for disease prediction, the proposed model can predict 23 diseases. For collecting the dataset, 40 people were examined, and five images were collected from each person, with a total of 200 images; 15 images were for each of the 24 disorders, and 40 images of healthy nails were taken. The proposed model produced average specificity, sensitivity, and accuracy values of 75.9%, 97.4%, and 89.9%, respectively. With the same number of predicted diseases, In Coimbatore, India, Maniyan and Shivakumar [12] proposed the system (NIPS-K), which gathers 13 features from the human nail, combining nail color, shape, and texture, and uses them to predict diseases. The dataset consists of 483 samples of palm pictures captured with a digital camera of 40 different individuals. Out of the 483 picture samples, 123 samples of healthy individuals and 15 samples of each of the 24 diseases were collected. Further, 13 features representing 24 diseases and a healthy nail were used in the training dataset.

Additionally, 240 pictures were used for training, and 120 pictures were used for the testing dataset out of the 360 total diseased pictures. The input to the system is the palm's back, which is taken by a camera. After that, the nail area's Region of Interest is automatically segmented. Then, the features of the nail are extracted and combined to create a feature vector. The K Nearest Neighbor (KNN) Classification Method is then used to categorize and predict the diseases. The model identified 24 diseases with an accuracy of 93.6388%. Other researchers in Thailand, Hongboonmee and Kanyaprasit [13] proposed a system that can use a smartphone camera to capture an image of the nail, analyze it, and categorize it using trained DL models. The system will next examine the set of diseases it has identified and show the results and the accuracy on the application screen. Rheumatoid, psoriasis, melanoma, and anemia are the five nail diseases represented by the 400 photos in the dataset. 20% of the data are used for testing with 80 images, while 80% are used for training with 320 images from the dataset. The data was collected using smartphone cameras in the region of Phitsanulok. Three models were developed and trained on the sample data using a Deep Convolutional Neural Network.

The experimental results indicate that MobileNet 1.0 achieved the highest precision and accuracy of 90.20%, compared to MobileNet 0.50 and MobileNet 0.75. Similarly, the idea of creating a mobile app as an interface attracted the interest of other authors. For example, In Mumbai, India, Mehra et al. [14] proposed a deep learning model to create a mobile app that can recognize diseases using uploaded nail photos. The concept of transfer learning was applied using a VGG16 pre-trained model by Keras framework with two additional layers of Artificial Neural Networks (ANN). They have identified nail diseases using the VGG16 model and classified them using CNNs. Additionally, they used Android Studio, Firebase API, and tflite to create an android application for the mobile app. The model was trained to classify three classes: subungual melanoma, yellow nail syndrome, and healthy nails. The dataset is gathered from different resources such as dermatologists and skin specialists

as well as many websites like dermnet, google images, and Kaggle. The dataset consists of three classes: melanoma with 192 images, yellow nail with 248 images, and a healthy nail class with 200 images; 67% of the dataset is used for training, and 33% for testing. The average accuracy was 94%. Furthermore, in Taipei, Taiwan, Liu et al. [15] developed a DL algorithm mask Region-based Convolutional Neural Network (R-CNN) to segment the nail disease pictures and categorize the lesions into 17 classes.

An instance segmentation algorithm known as the Mask R-CNN model examines the entire picture and identifies each feature separately. The dataset contains 17 classes, including alopecia areata, clubbing, darters disease, eczema, koilonychia, leukonychia, onycholysis, pale nail, red lunula, splinter haemorrhage, terry's nail, white nail, and yellow nails. The dataset contained 655 photos; 80% for training with 524 images, 10% for validation with 66 images, and 10% for testing with 65. In order to enhance the model's ability to generalize to different sets of imaging conditions, data augmentation was applied using pre-processing techniques such as reverse upside down, left and right, rotating -45 to +45 degrees, rotating -20 to +20 degrees, and scaling 0.5 and 1.5 times. The classification precision percentage for the model was 87.69%. Moreover, In Pune India, Thakare et al. [16] have proposed an automated medical support system for detecting human health based on the Bicluster method for nail color analysis. The dataset was collected via medical professionals and other helpful websites, like Dermnet and Medicinenet. The system is capable of detecting seven illnesses based on the color and texture of the nail, including diabetes, heart disease, liver disease, psoriasis, cancer, and leukemia. Moreover, Support vector machines, neural networks, Multilayer Perceptron (MLP) neural networks, and decision tree algorithms were employed as classification methods. The Weka tool was used to implement the decision tree generator. The RGB colors in the nail color model are extracted by leaving 1/6th margins vertically and horizontally to retrieve the real average of pixels that enter the faulty region and give the classification algorithm the features it needs. The texture feature was computed using the Gray-level co-occurrence matrix (GLCM) technique. The model has achieved an accuracy of 86.33% using the MLP neural network by combining the AverageColor, Bicluster, and GLCM methods. Likewise, by predicting based on the nail color, In Tamil Nadu, India, the model proposed by K. Dhana Shree. et al. [17] use a high-resolution digital camera or scanner to take a picture of a fingernail.

These photos will then be divided into various parts using image segmentation algorithms until the area of interest is found. Following the extraction of the features, the disease will be predicted using the color and texture of the nail as a guide. To improve the effectiveness of data categorization, the system employs a number of ML algorithms, including SVM, Multiclass SVM, Ant Colony Optimization (ACO), and CNN. The Messidor and Kaggle websites provided the dataset for this suggested system. The method can identify six different disorders, including melanonychia, Terry's nails, and onychogryphosis. Precision, F1 Score, sensitivity, and accuracy are the criteria that are used to evaluate performance. Precision, F1 Score, Sensitivity, and Accuracy for the Multiclass SVM are 89%, 86%, 88%, and 88%, respectively, while for the CNN, the values are 93%, 87%, 89%, and 94.673%, respectively. According to the results, CNN is better than Multiclass SVM at detecting the disease from nail

images. In this work an integrated DHL approach is investigated for nail disease classification. This approach combines DL pre-trained models for feature extraction with ML classifiers. Moreover, it is suited for small datasets, performs auto-feature extraction, and requires less computational power since it leverages the advantages of traditional DL and ML while alleviating the drawbacks of both techniques.

III. METHODOLOGY

A. Convolutional Neural Network (CNN)

CNNs are known for their ability to efficiently identify objects and learn highly abstract features. The architecture of a general CNN model is made up of four parts: (a) convolution layer, (b) pooling layer, (c) fully connected layer, and (d) activation function. The functionality of each part is described briefly below [18].

1) Convolution Layer

The input layer receives an image to be classified, and the output is the predicted class label computed using image-extracted features. A receptive field is formed when an individual neuron in the next layer is connected to some neurons in the previous layer. The receptive field is used to extract local features from the input image. To generate the feature map, the weight vector is slid over the input vector. Convolution operation refers to the method of sliding the filter horizontally and vertically.

2) Pooling Layer

The main benefit of employing this layer is that it significantly decreases the number of trainable parameters while also introducing translation invariance, indicating that even if the inputs is converted, the CNN will still recognize the class of this input. To execute pooling, a window is chosen, and the input items within that window are processed via a pooling function which generates another output factor. There are many pooling techniques, such as (Max-Pooling and Average-Pooling, etc.) but Max-Pooling is widely used as it significantly reduces the map size.

3) Fully Connected Layer

The result of the first phase, which comprises convolution and pooling, is transferred into the fully connected layer. Finally, the dot product of the weight vector and the input vector are calculated to give the final output. Gradient descent is a learning algorithm used to decrease the cost function by calculating the cost over a whole training dataset and updating the parameters only after one epoch, where an epoch means to comb the entire dataset single time forward and backward.

4) Activation Function

Activation functions are used to add non-linearity to the network. There are many types of activation functions such as the sigmoid activation function which is widely used in traditional machine-learning techniques. Also Rectified Linear Unit (ReLU) has proven to be more effective in introducing non-linearity. Moreover, the Softmax function which is used to transfer the numbers to probabilities and these probabilities represent the prediction of the classes the model will use.

B. Machine Learning Classifiers

Logistic Regression Estimates discrete values, which are dependable categorical variables, such as: yes and no, but mainly binary numbers 0 and 1. The idea of the logistic

regression is the final outcome will not directly be 0 or 1, but probabilistic values between them [19]; to predict an event probability by machine language fitting function. It's a significant machine language algorithm shaped as the letter "S" shown in Fig. 1, which helps in having the ability to distinguish the major differences in the model, such as: is the image classifies as a healthy nail or not; if not, is it melanoma or beau's lines, until reaching the final outcome.

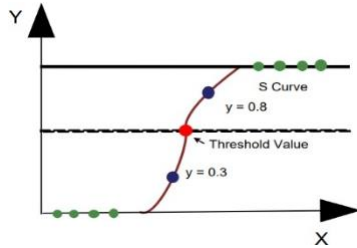


Fig. 1. Logistic Regression structure.

Support Vector Machine (SVM) can be used to identify and classify extreme levels of decision classifications, such as face reading, categorizing texts, and image processing which is our case. The SVM algorithm identifies the extreme points that are used while creating the hyperplane shown in Fig. 2, which is a decision boundary that distinguishes the SVM classes, these points are said to be support vectors. The distance between the support vector and the hyperplane must be at maximum level without any interference otherwise, it will change the structure of the hyperplane [20].

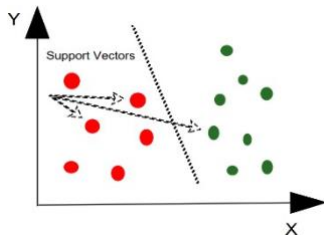


Fig. 2. SVM Hyperplane

Stochastic Gradient Descent (SGD) is an optimization technique to train models, and it is capable of finding optimal solutions to the large range of problems. It is fitted into two arrays: array x that hold train samples and array y that hold class labels. After fitting the model new values are predicted. The hyperplane in gradient descent is controlled by the parameter fit_intercept. The problem here is presented as a regression problem. Therefore, the predicted class will correspond to the sign of the y array which is the target. The primary idea is to tweak parameters iteratively in order to minimize the cost function; the focus will be on the steps known by the rate of hyperparameters, and determine if it's too small or too high, in order to identify the optimal solution value [21]. And of course, making sure that all functionalities have similar scale otherwise the converge time will be stagnant. Batch is total number of randomly selected samples that is used for calculating the gradient for each iteration. the equation to calculate SGD therefore is (1):

$$\text{for } i \text{ in range } (m):$$

$$\theta_j = \theta_j - \alpha(\hat{y} - y^i)x_j^i \quad (1)$$

C. Dataset Description

In this study, Due to the limited image sources of beau's lines, eczema, and melanoma diseases, the dataset was gathered from different sources, and it consists of 821 images divided into training and testing, illustrated in Table I. The training dataset comprises 75% of the total images divided into 155 beau's Lines images, 152 eczema images, 154 melanoma images, and 156 healthy nails images. The testing dataset comprises 25% of the total images divided into 52 beau's lines images, 50 eczema images, 50 melanoma images, and 52 healthy nails images. Fig.3 displays some samples of the dataset.

Table I: Dataset Division

Train Dataset (75%)	Total Number of Images	Test Dataset (25%)	Total Number of Images
Beau's Lines	155	Beau's Lines	52
Eczema	152	Eczema	50
Melanoma	154	Melanoma	50
Healthy Nails	156	Healthy Nails	52
Total	821		



Fig 3. dataset samples

IV. PROPOSED MODEL ARCHITECTURE

One of the lengthiest stages in machine learning is feature engineering, which is the manipulation of features in the dataset to obtain the best performance. It is a challenging but necessary step when training a machine learning model. On the other hand, one of the major benefits of deep learning is that it eliminates the need for manual feature engineering; instead, the deep learning model scans the dataset, identifies significant features, and uses them to train the model. However, the vast amount of data required to train a deep learning model puts it at a disadvantage compared to traditional machine learning techniques.

Considering the pros and cons of both approaches and the limited size of our dataset, the proposed DHL will take the best of both worlds. This approach combines DL and ML methods, resulting in a hybrid structure that will highlight the benefits of both methods and provide an accurate and computationally efficient solution. The DL models extract features from the data without any manual intervention, while the traditional ML will be used to classify the data

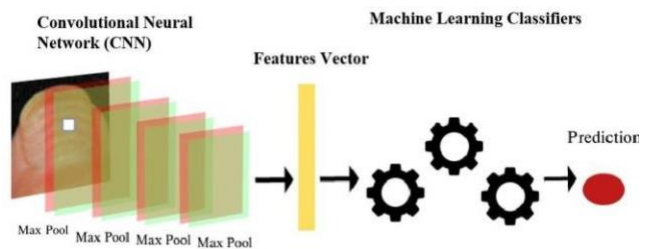


Fig. 4. The Flow Diagram of The Model

Fig 4 illustrates the structure of the proposed hybrid model. The first part of the model is a pre-trained CNN such as VGG16, InceptionV3, DenseNet201, etc., which receives the preprocessed nail image and sends it through different layers to extract features. Since the goal is to use the CNN as a feature extractor, not a classifier, the last fully-connected layer is removed and all layers were frozen to prevent the weights from updating. The purpose of using a pre-trained network to extract features instead of building a Deep Neural Network from scratch is the fact that using a pre-trained network is more efficient with smaller datasets [22]. The output of the CNN is a feature vector that is reshaped to be an input to an ML classifier.

Below is an algorithm that further demonstrates the overall process for the DHL structure. First, the input image is preprocessed by resizing it to a uniform size, in our case 224×224 , then the image is normalized by dividing its pixels by 0.255; this step ensures that the pixels range from [0,1] instead of [0,255], which will provide faster and easier computation. The final step in image preprocessing is expanding the dimension of the image to 4 dimensions (number of samples, height, width, channels). Once preprocessing is complete, the image is fed to a CNN model, which will return a feature vector of the image. Finally, the feature vector is reshaped and sent to the ML classifier, which will return the predicted class.

ALGORITHM 1: DEEP HYBRID LEARNING ALGORITHM	
	Input: Image
	Output: The predicted class
	#Image Preprocessing
1	Image ← Resize image // 224×224
2	Image ← Normalize image // divide by 0.255
3	Image ← Expand dimensions // (n_samples, height, width, channels)
4	#Feature Extraction Using CNN
5	Features ← CNN_Model (Image)
6	Features ← Reshape the Image to send to ML_classifier() // (n_samples, n_features)
7	#Classification Using ML classifier
8	Prediction ← ML_classifier (Features)
9	Print (Prediction)

V. PERFORMANCE EVALUATION METRICS

Performance evaluation is an essential step to guarantee the efficiency of the model. The performance of the model will be evaluated using the training set which consist of around one hundred and fifty images for each of the four classes. The performance analysis was done using the statistical measures for classification like Precision, Recall, F1-score and Accuracy. Obtaining such measures is straightforward in the case of binary classification. However, a multiclass problem requires a different approach when evaluating the model since

the evaluation metrics have to be calculated for each class. Then the total for all classes is considered. Below is the confusion matrix in the case of a multiclass classification problem.

TABLE II. Multiclass Confusion Matrix

ACTUAL	PREDICTED				
	CLASSES	A	B	C	D
	A	TP _A	E _{AB}	E _{AC}	E _{AD}
	B	E _{BA}	TP _B	E _{BC}	E _{BD}
	C	E _{CA}	E _{CB}	TP _C	E _{CD}
	D	E _{DA}	E _{DB}	E _{DC}	TP _D

The letters A, B, C, and D represent different classes predicted by the model. The diagonal cells denoted as TP_{class} represent the True Positive (TP) values. Furthermore, the cells denoted as E_{XY} represent X values that were erroneously predicted as Y. Using the above confusion matrix the following evaluation metrics can be calculated:

- I. **Accuracy:** the ratio of correct prediction over all predictions made by the model.

$$Accuracy = \frac{\text{Correct Predictions}}{\text{All Predictions}} \quad (2)$$

- II. **Precision:** it is the total number of true positives for a certain class divided by the true positive of the whole class (column). For example, to calculate the precision of class A in Table II.

$$Precision A = \frac{TP_A}{TP_A + E_{BA} + E_{CA} + E_{DA}} \quad (3)$$

- III. **Recall (sensitivity):** it is the total number of true positives for a certain class divided by the true positive of the whole prediction of a certain data (row). For example, to calculate the precision of class A in Table II.

$$Recall A = \frac{TP_A}{TP_A + E_{AB} + E_{AC} + E_{AD}} \quad (4)$$

- IV. **F1-score:** it is the harmonic mean of precision and recall. In the case of multiclass classification, it is calculated using the precision and recall of a certain class. For example, to calculate the f1-score of class A in Table II.

$$F1 - score A = 2 \times \frac{\text{Precision}_A \times \text{Recall}_A}{\text{Precision}_A + \text{Recall}_A} \quad (5)$$

The evaluation metrics for each class is obtained. However, only a single value for each metrics is needed to represent the model's performance. Therefore, one of the three averaging techniques utilized in multiclass problems will be used to calculate the final value of each metric.

- I. **Macro:** it is the mean of all metrics across classes, it assigns equal weights for all classes which makes it suitable for balanced problems.
- II. **Weighted:** consider class imbalance by multiplying the evaluation metric score by the number of occurrences of each class.
- III. **Micro:** it is identical to accuracy, it is calculated by dividing the sum of the diagonal cells by the sum of all cells.

VI. EXPERIMENT & RESULTS

The proposed model is built using python. Furthermore, various libraries such as Tensorflow, Scikit-learn, and more were utilized to pre-process the images and train and test the model. Two approaches were tested to classify nail diseases. The first approach is transfer learning, where a pre-trained model is fine-tuned and used to classify a specific dataset. The second approach is DHL, where a DL model is used for feature extraction, and a traditional ML Classifier is used for classification. The results for both experiments are recorded and compared to show the superior approach.

A. First Approach: Transfer Learning

Observing the limited size of our dataset, experimenting with the transfer learning approach was the sensible option. Transfer learning is typically used for tasks where the dataset has limited data to train a full-scale model from scratch. A couple of CNN pre-trained models (VGG16, Inceptionv3, DenseNet201, etc.) were trained on our nail diseases dataset. The ImageDataGenerator class is utilized for all models to generate more train/test images. Also, all the trained CNN weights layers were frozen to prevent the weights from updating during the training process and to prevent distorting any of the information contained inside them during future training process, and trained the new four layers on the dataset.

Table III. Evaluation of CNN models performance

Model	Precision	Recall	F1-Score	Accuracy
VGG16	90%	90%	90%	90%
Inception V3	86%	84%	85%	84%
DenseNet 201	81%	80%	79%	80%

Table III above shows the evaluation of CNN models performance. All three models were trained for 150 epochs. The VGG16 demonstrated the best performance out of the three models with an accuracy, precision, recall, and f1-score of 90%. Observing the evaluation metrics of the three models, Obviously, the performance of the models is satisfactory relative to the small size of the dataset used for training. However, to explore the possibility of obtaining higher results with the proposed DHL approach, the three pre-trained CNN models were implemented to extract features from the images (part B).

B. Proposed Approach: Deep Hybrid Learning (DHL)

The proposed approach will utilize the convolutional filters, which are a part of the CNNs, to extract features from an input image and use these features as input to a traditional ML classifier. To achieve this, it is important to note the following: Since the primary aim is to apply CNNs for feature extraction only, removing the fully connected (Dense) layers responsible for classification is crucial. Furthermore, all models use the infamous ImageNet [23], a dataset with over 14 million images belonging to 1000 classes, to instantiate the model's weights. Therefore, all layers of the CNN model are frozen to prevent the weights form updating. Finally, the output of the CNN model will be a feature vector with the shape (n_samples, height, width, channels). However, the ML classifier is expecting an input of shape (n_samples,

n_features). Therefore, the feature vector must be reshaped before being fed to the ML classifier.

Observing the results of the transfer learning approach, the three best-performing DL models were VGG16, InceptionV3, and DenseNet201. Therefore, in the following sections, these models will be experimented through combining them with different ML classifiers and recording the performance metrics, including the accuracy, precision, recall, f1-score, and execution time.

VGG16 For Features Extraction

With the transfer learning approach, the VGG16 exhibited the best performance out of the three models, with an accuracy of 90%. Hence, with the DHL approach, the VGG16 will be used for feature extraction only and combined with three ML classifiers, namely, LogisticRegression, LinearSVC, and SGDClassifier, to find the best-performing combination.

TABLE III. VGG16 Accuracy Assessment

Feature Extractor	Classifier	Precision	Recall	F1-Score	Accuracy
VGG16	LogisticRegression	84%	83%	83%	82%
VGG16	LinearSVC	84%	82%	82%	82%
VGG16	SGDClassifier	74%	70%	68%	69%

Table IV shows that the best performance is achieved when the VGG16 model is combined with the LogisticRegression classifier as it reached an accuracy of 82%, a 13% improvement in accuracy compared to when combined with the SGDClassifier. However, compared to the first approach, "Transfer Learning", the model's accuracy decreased from 90% to 82%.

InceptionV3 For Features Extraction

In the first approach, InceptionV3 came in second after the VGG16 model with an accuracy of 84%. So, again, the InceptionV3 will be used for feature extraction and combined with the same three Machine Learning classifiers used above.

TABLE IV. InceptionV3 Accuracy Assessment

Feature Extractor	Classifier	Precision	Recall	F1-Score	Accuracy
InceptionV3	LogisticRegression	89%	87%	87%	87%
InceptionV3	LinearSVC	88%	87%	87%	87%
InceptionV3	SGDClassifier	86.7%	85%	85%	84.7%

Table V shows that the InceptionV3 model performs better than the VGG16 model with the DHL approach. It reached an accuracy of 87%, an increase of 5% compared to the best accuracy achieved by the VGG16, which is 82%. Furthermore, the accuracy seems to have enhanced compared to the transfer learning approach, where it reached an accuracy of 84%.

DenseNet201 For Feature Extraction

The Densenet201 came in last in the initial experiment compared to VGG16 and InceptionV3 with an accuracy of 80%. However, this does not necessarily mean that its performance can not be improved using the DHL approach, as seen with the InceptionV3. Therefore, again, the Densenet201 will be used for feature extraction and combined with the same three ML classifiers used above. Despite coming last with the transfer learning approach experiment, the above table shows the remarkable enhancement in the DenseNet201 model's accuracy with the DHL approach.

TABLE V. DenseNet201 Accuracy Assessment

Feature Extractor	Classifier	Precision	Recall	F1-Score	Accuracy
DenseNet201	SGDClassifier	94%	94%	94%	94%
DenseNet201	LinearSVC	91%	91%	91%	91%
DenseNet201	LogisticRegression	90%	90%	90%	90%

It reached an accuracy of 94%, an increase of 14% compared to the accuracy with the transfer learning approach.

Execution Time Assessment

Occasionally, a model's run time and accuracy are trade-offs since it may be reasonable to sacrifice some accuracy in favor of a shorter run time, especially in situations where a long waiting time is not acceptable, such as medical diagnosis. Therefore, to specify the optimal model architecture for our nail diseases dataset, each model is trained and tested three times, and the following is calculated:

- Average training time: the time taken to extract features from the training dataset and fit the model.
- Average testing time: the time taken by the model to extract features and predict the classes on the testing and validation dataset.
- Average prediction time: the time taken by the model to extract the features and predict the disease on a single image.

It is worth mentioning that the execution time of an algorithm is affected by the dataset type and size. Therefore, the results may vary when running the same experiment on datasets with different sizes or data types. However, the results will help us understand which feature extraction and classification algorithms combination are optimal for our case.

TABLE VI. Execution Time Assessment

Feature Extractor	Classifier	Avg Training Time (s)	Avg Testing Time (s)	Avg Prediction Time (s)
DenseNet201	SGDClassifier	19.3	3	0.6
DenseNet201	LinearSVC	48.6	5	1.3
DenseNet201	LogisticRegression	114.3	3.6	2.6
InceptionV3	SGDClassifier	10.7	1	0.7

InceptionV3	LinearSVC	17	1.3	0.7
InceptionV3	LogisticRegression	84.7	1.7	0.7
VGG16	SGDClassifier	18	3	1
VGG16	LinearSVC	39	4	1
VGG16	LogisticRegression	76.3	4	1

Table VII shows that the DenseNet201 exhibits the best performance when combined with the SGDClassifier. It is noticeably better than the other combinations in the prediction time. The DenseNet201 combined with the LogisticRegression classifier was the slowest, with an average of 2.6 seconds to perform feature extraction and classification on a single image, taking 2 more seconds compared to the fastest model (DenseNet201 and SGDClassifier). In all cases, the SGDClassifier was the most efficient in terms of execution time. It had the fastest training and testing time when combined with the InceptionV3 and the fastest prediction time when combined with the DenseNet201.

The experiment above indicates that the DHL approach outperforms the transfer learning approach as it managed to reach an accuracy 4% higher than the one achieved with transfer learning. The best DHL model architecture in terms of accuracy and prediction time is the DenseNet201 for feature extraction and the SGDClassifier for classification, as it reached an accuracy of 94% and an average of 0.6 seconds for prediction time. The classification report, and the confusion matrix of the model are displayed in Fig 5, and Fig 6.

	precision	recall	f1-score	support
Beau_s Lines	0.91	1.00	0.95	52
Eczema	0.92	0.92	0.92	50
Healthy Nails	1.00	0.85	0.92	52
Melanoma	0.92	0.98	0.95	50
accuracy			0.94	204
macro avg	0.94	0.94	0.94	204
weighted avg	0.94	0.94	0.94	204

Fig 5. Classification Report

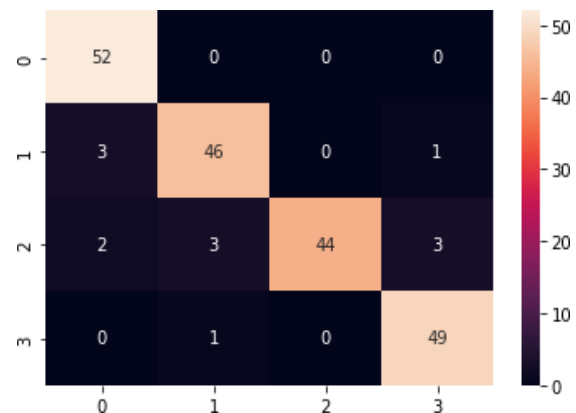


Fig 6. Confusion Matrix

Furthermore, some predictions made by the model on the test set are displayed in Fig. 7, Fig. 8, Fig. 9, and Fig. 10.

The prediction for this image is: ['Beau's Lines']
The actual label for this image is: Beau's Lines



Fig 7. Test Case 1

The prediction for this image is: ['Eczema']
The actual label for this image is: Eczema

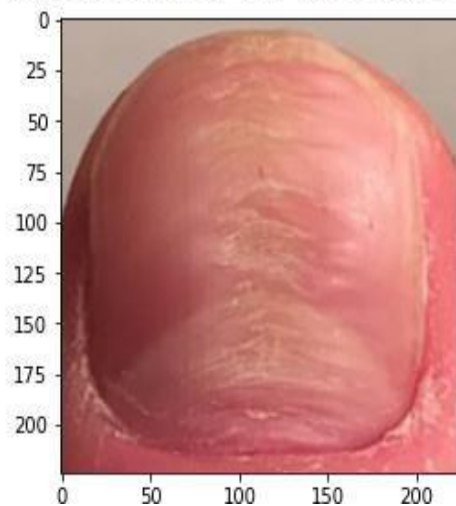


Fig 8. Test Case 2

The prediction for this image is: ['Melanoma']
The actual label for this image is: Melanoma

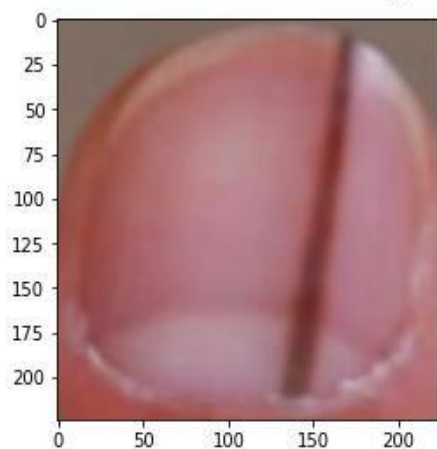


Fig 9. Test Case 3

The prediction for this image is: ['Healthy Nails']
The actual label for this image is: Healthy Nails

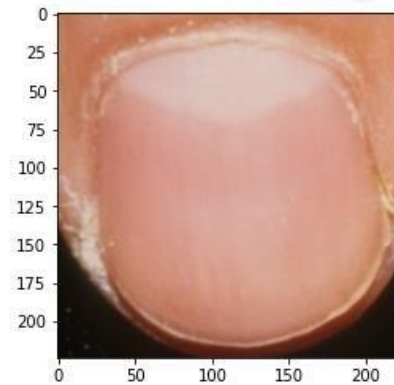


Fig 10. Test Case 4

TABLE VII. Comparison With Previous Studies

Study	Method	No. of classes	Accuracy
This study	DHL	4	94%
Maniyan and Shivakumar [11]	One-vs-Rest for SVM	23	89.9%
Nijhawan et al. [10]	Ensemble of CNNs	11	84.58%
Indi et al. [8]	RGB analysis	5	65%

Table VIII compares some of the studies in the literature review with the proposed approach. The proposed DHL approach achieved the best results with an accuracy of 94%, an increase of 29% in accuracy compared to the study conducted by Indi et al. [8], where the nail color is the only feature, and 9.42%, 4.1% enhancement in accuracy compared to the studies conducted by Nijhawan et al. [10], and Maniyan and Shivakumar [11] respectively.

VII. CONCLUSION

This work implements a Deep Hybrid Learning approach by using a pre-trained Convolutional Neural Network to extract features, and Machine Learning algorithm was used for classification. The Algorithms of CNNs applied were VGG16, InceptionV3 and DenseNet201. The classification algorithms applied were Logistic regression, Support Vector Machine and Stochastic Gradient Descent.

This proposed system can diagnose four categories of nail conditions, namely beau's lines, melanoma and eczema, and healthy nail conditions. And it gives more accurate results than human vision and minimizes the cost of the diagnosis in hospitals. In the experimental evaluation, DenseNet201 combined with the Stochastic Gradient Descent Classifier reached the highest performance out all the models with 94% precision, 94% recall, 94% F1-Score, and 94% accuracy.

In the future, the proposed model could be enhanced to take textual input of the patient's symptoms along with nail images for a more accurate diagnosis. Also, more nail diseases can be predicted by enlarging the dataset.

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